



COMP 7940 Cloud Computing Project Presentation

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Project Overview: Telegram bot “Maestro”

The problem:

Students find it hard to form study groups with people who share the same hobbies or speak the same language.

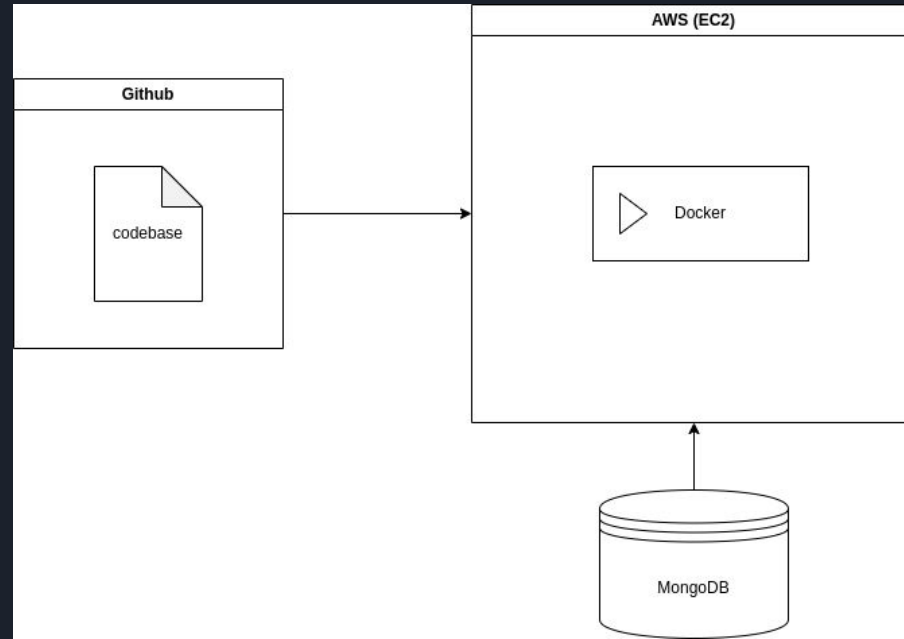
Students can have simple questions that don't know where to find the answers about the course or the project

The solution:

Students now can use the Telegram bot to search for people that have similar language and interest to communicate better in a group project. Also with the telegram bot students can now ask simple questions for course information.

Architecture and Cloud Design

Simple, practical, fully cloud-native workflow





DevOps Practices

Tech Stack: Python + MongoDB Atlas

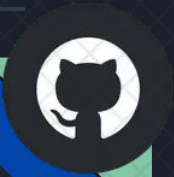
Orchestration of multiple containers: Docker Compose
(bot + Celery workers + Redis + Flower)

Task management: Celery + Redis + Flower

Version Control: Git repository on GitHub

CI/CD: GitHub Actions → AWS EC2 Instance

Security: GitHub Secrets + *.gitignore*



GitHub

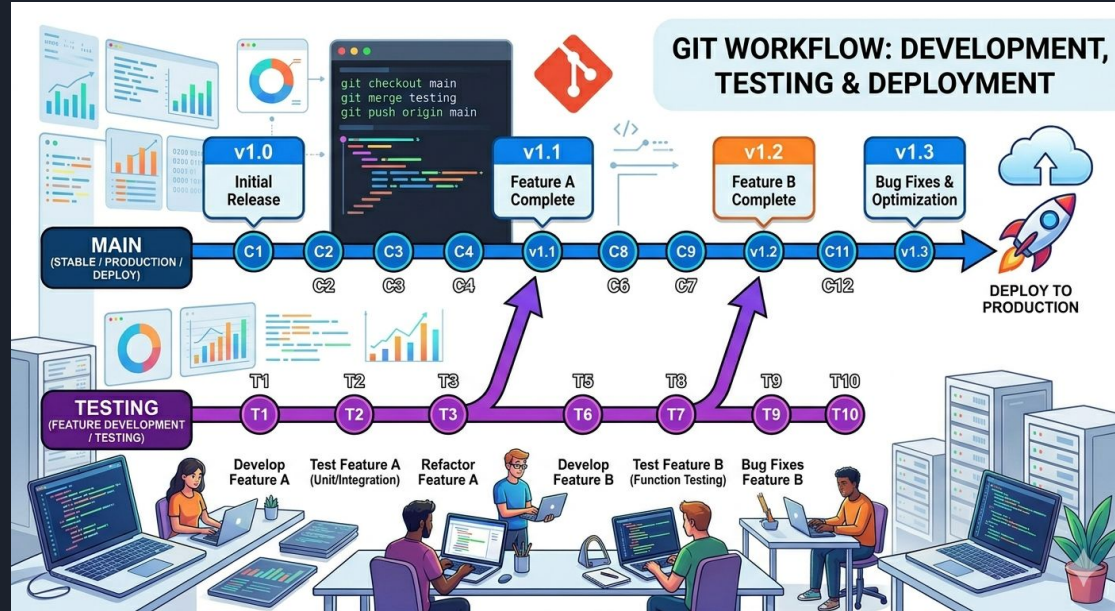


Image generated with Nano Banana 2

MongoDB Atlas

ORGANIZATION
Sylwester's Org - 202...

PROJECT
Cloud

IP Access List

! You will only be able to connect to your cluster from the following list of IP Addresses:

IP Address	Comment	Status
13.211.24.17/32	EC2	Active

MongoDB Compass - chatbot.nejdxh9.mongodb.net/chatbot_db.users

Connections Edit View Collection Help

Compass

() My Queries

CONNECTIONS (5)

Search connections

- bigdata.mweq54.mongodb.net
- chatbot.nejdxh9.mongodb.net
 - admin
 - chatbot_db
 - groups
 - users
 - config
 - local
 - flask_db
 - hiddenbot
 - kioskdb

chatbot.nejdxh9.mongodb.net > chatbot_db > users

Documents 53 Aggregations Schema Indexes 3 Validation

{ language: "English" }

Generate query Explain Reset Find Options

ADD DATA EXPORT DATA UPDATE DELETE INSIGHT

100 1 - 36 of 36

username: "lasagnalegend_47"

_id: ObjectId('69dcb8df13fa9e19ba2edbb1')
telegram_user_id: 900948
age: "21-25"
completed: true
created_at: 2026-04-13T09:34:05.895+00:00
display_name: "Polenta Phantom"
gender: "Male"
hobbies: Array (3)
fn_group: true
language: "English"
updated_at: 2026-04-13T18:45:59.897+00:00
username: "polentaphantom_48"

_id: ObjectId('69dd1ef4f13fa9e19ba2ee58')
telegram_user_id: 711718736
completed: true
created_at: 2026-04-13T16:51:00.850+00:00
messages: Array (2)
updated_at: 2026-04-14T08:31:43.002+00:00
username: "Sylwe17"
age: "Under 18"
gender: "Male"
hobbies: Array (2)
fn_group: false
language: "English"

ORGANIZATION
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PROJECT
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Database Users

User	Description	Authentication Method	MongoDB Roles	Resources
Gino		SCRAM	atlasAdmin@admin	All Resources

Monitoring and Observability

Flower WebUI + AWS CloudWatch

Real-time task monitoring + AWS resource metrics

Flower Workers Tasks Broker Documentation

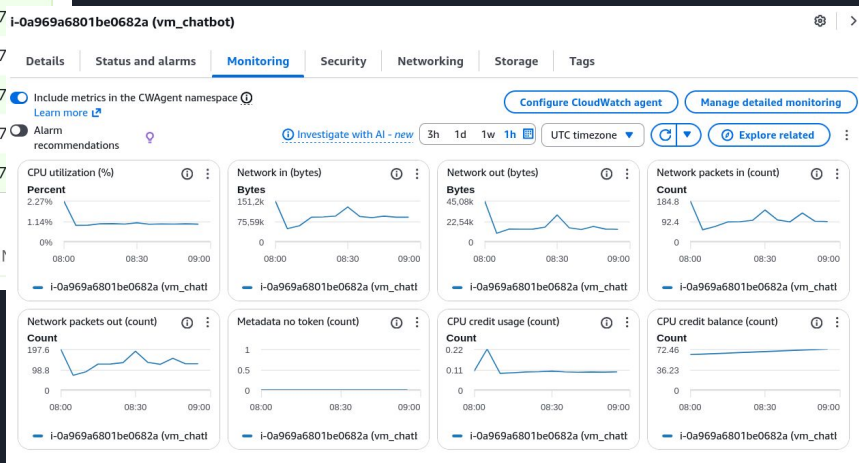
Show 15 workers

Search:

Worker	Status	Active	Processed	Failed	Succeeded	Retried	Load Average
celery@2d6ea0be78ea	Online	0	0	0	0	0	5.5, 80.41, 7
celery@a3237ad087fd	Online	0	0	0	0	0	5.5, 80.41, 7
celery@afd07d968b0b	Online	0	0	0	0	0	5.5, 80.41, 7
celery@16d55cebfcd	Online	0	0	0	0	0	5.06, 79.08, 7
celery@b4782b0cc6ec	Online	0	0	0	0	0	5.5, 80.41, 7
Total		0	0	0	0	0	

Showing 1 to 5 of 5 workers

Previous 1





LLM Integration (Virtual Professor)

Integrates the HKBU GenAI LLM API with clear prompt templates and a Markdown file called *course_info.md*, containing all the necessary information for a natural-language course Q&A.



Cost Awareness and Optimisation

Current setup = completely free (free tiers everywhere)

Estimated monthly cost if scaled:

AWS t3.medium $\approx$ \$30

MongoDB M10 $\approx$ \$57

Redis + LLM $\approx$ \$40–100

Total $\approx$ \$130–270 (with savings plans possible)

As it stands, the current setup is very cost-aware



Job Division and Team Collaboration

Sylwester Krasinski: Managed the MongoDB Atlas cloud database integration and persistence of data.

Alessandro Di Fabio: Oversaw the testing protocols, logic validation and overall project planning.

Amedeo Di Biase: Managed the DevOps workflow, container orchestration and CI/CD pipeline.



Conclusion and future outlook

Result: A modular, scalable chatbot that successfully bridge the gap between students and the requests for group projects

Future work:

- Scaling to multiple university courses
- Advanced sentiment analysis for student feedback
- Integration with official university calendars



Thanks for your attention